TUSHAR LACHMAN

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PROFESSIONAL SUMMARY

Computer Science student at RMIT University, minoring in Artificial Intelligence & Machine Learning, who ships production software end to end. Built and sold a medical practice management platform now running a Jakarta family practice of 1,000+ patients, plus five further products across web, mobile, native Android and Python — several with real users, all integrating third-party and AI APIs, all documented at tusharlachman.dev. Seeking a summer internship (Nov 2026 – Feb 2027) or a graduate role from mid-2027, open to any technical role.

Work rights: Student visa (subclass 500) — unrestricted hours during course breaks, 48 hours per fortnight in session; eligible for a Temporary Graduate visa (subclass 485) from August 2027.

PROJECTS

Medical Practice Management Platform
June 2026 – Present

React · TypeScript · Supabase · Deno · Capacitor · Twilio — private repo, tusharlachman.dev/work/medical

- A TypeScript monorepo on npm workspaces — React dashboard, two Android apps (Capacitor), shared document templates — behind 89 Deno edge functions and 54 migrations, covered by 1,975 unit tests.
- Rebuilt WhatsApp booking as a 15-tool Claude agent that patients book, move and pay through by tapping, in English or Bahasa — every change gated in code on the patient's yes, tested in 48 real-model conversations.
- Audited the live database with nothing but the read-only anon key that ships in the client bundle: eleven tables still answered, because row-level security sat switched off beneath correct-looking policies. Also closed a public OAuth callback that could have sent the nightly backup to a stranger's Drive.
- Insurers' own claim forms filled in their own boxes, tried on 16 real insurers' forms; a 40-tool Claude assistant that always proposes and never commits; every Claude request de-identified to a patient number.

FitScroll — Earn-Your-Screen-Time Android App
August 2026

Kotlin · Jetpack Compose · CameraX · ML Kit · Accessibility Services — github.com/TusharLachman25/FitScroll

- The camera counts push-ups on-device, banks each rep as a minute of screen time, and locks the apps you pick at zero. Minutes expire 24 hours after they are earned.
- ML Kit pose detection behind a 1–5 strictness dial. A rep also needs the shoulders to travel a fraction of the user's own shoulder-to-hip length — without it, standing up and curling your arms passes every angle gate there is.
- 125 unit tests, no device needed: bank, counter, blocking decision and drain meter are pure functions taking an explicit clock. The lock is an accessibility overlay, so it lands on a running app mid-scroll.

Jarvis — Personal Life OS
June 2026

Next.js 16 · React 19 · TypeScript · Supabase · Claude API · Google APIs — github.com/TusharLachman25/jarvis-life-os

- Tasks, calendar, health and finance in one dashboard: 38 REST endpoints, Google OAuth with token refresh across Calendar, Tasks and Gmail, and Claude parsing bank statements into reviewable transactions.
- Reconciled two wearable ecosystems that do not interoperate — Fitbit through Google Health's live API, Galaxy Watch through a Samsung Health export Claude parses — keyed by source id, so a re-import is a no-op.

Locked In — Social Fitness App
April – May 2026

React Native (Expo) · TypeScript · Supabase · Gemini API · EAS Build — github.com/TusharLachman25/Locked-In

- A squad-based workout tracker in daily use, shipped as an Android APK with a Vercel web build. Stories, leaderboards, a feed and messaging on Supabase, with push sent by a Postgres trigger.
- Gemini vision reads a smartwatch screenshot and auto-fills activity, distance, duration and calories.

Kitchen OS — Pantry & Nutrition Assistant
February – March 2026

Python · Streamlit · Supabase · Google Gemini · pandas · Altair — github.com/TusharLachman25/kitchen-os

- A pantry, recipe and nutrition tracker in Python and Streamlit over a 66-function data layer.

- Gemini drives a conversational recipe assistant, and pantry stock is deducted when a recipe is cooked.

Other Projects 2026

- **Bets — The Vault** · React Native, Supabase — a shared ledger for informal bets, from one codebase.
- **This portfolio** · Next.js 16, React 19 on Vercel — tusharlachman.dev, documented screen by screen.

Academic Projects — RMIT University 2024 – 2026
Python · PyTorch · scikit-learn · Flask · Docker · AWS · C · C++ · Java · NetworkX · transformers · Clingo · Prolog

- **Machine learning:** a CNN over 20,280 colon-tissue patches against logistic-regression, random-forest and SVM baselines, split by patient — 0.852 macro-F1 against a 0.90 target, reported as not met.
- **Cloud:** a Flask REST API on DynamoDB and S3 deployed three ways to compare — EC2 under gunicorn, ECS Fargate behind a load balancer, Lambda behind API Gateway.
- **Systems:** a multithreaded file copier in C (bounded queue, three mutexes, two condition variables, valgrind-clean) and an sbrk heap allocator; Clingo exam timetabling checked by an independent Prolog validator.
- **NLP, networks & security:** a six-stage YouTube pipeline (VADER/RoBERTa sentiment, LDA/BERTopic topics, NetworkX centrality, Louvain communities); a three-party GPG exchange and a default-deny iptables firewall in Kali.

EXPERIENCE

Independent Software Developer June 2026 – Present
Dr Varkha's Practice | Jakarta, Indonesia

- Sole developer of the platform above, on a paid monthly subscription: design, build, deploy and support.

Information Systems Intern August 2023 – January 2024
Makmur Jaya Abadi | Jakarta, Indonesia

- Digitised accounting records into Excel and Google Sheets, automating repetitive entry with formulas.

Part-Time Medical Assistant June 2020 – April 2023
Dr Varkha's Teleconsultations | Jakarta, Indonesia

- Digitised patient records, compiled laboratory and x-ray reports carefully protecting patient information, and supported staff by answering phones and booking appointments — the job the platform above replaced.

EDUCATION

Bachelor of Computer Science — Minor in Artificial Intelligence & Machine Learning Expected July 2027
RMIT University, Melbourne, Australia

Elementary School, Middle School and High School Completed April 2023
National High Jakarta School, Jakarta, Indonesia — Cambridge IGCSE, AS Level and A Level examinations completed.

Balvikas Student August 2011 – August 2023
Sathya Sai Education, Jakarta, Indonesia

SKILLS

Languages	Python, TypeScript, JavaScript, Kotlin, Java, C, C++, SQL, Prolog, HTML, CSS
Frameworks	React, React Native (Expo), Next.js, Node.js, Flask, Streamlit, Tailwind CSS, Capacitor
Mobile	Android (Kotlin, Jetpack Compose), CameraX, ML Kit pose detection, accessibility services
Cloud & DevOps	AWS (EC2, ECS, Lambda, S3, DynamoDB), Docker, Vercel, EAS Build, Gradle, Firebase, Git
Backend & Data	PostgreSQL, Supabase, REST APIs, serverless functions, OAuth 2.0, row-level security
AI & Data Science	Claude API (tool use), Google Gemini (text and vision), PyTorch, scikit-learn, pandas, NetworkX
Practice	Testing (Vitest, pytest, JUnit, Robolectric), monorepos, CI, technical writing

COMMUNITY SERVICE, INTERESTS & LANGUAGES

The Care Club — volunteer every 2–4 weeks with a charity providing food, groceries and medical care to families in need, alongside a GP, optician and dentist.

Languages: English (Proficient, C2) · Bahasa Indonesia (Advanced, C1) · Hindi and Mandarin (Beginner, A1)